

Statistics (AI and CS)

Week 2: Populations

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Reminder

Statistics: Use measurements on gathered *data* from a *sample* to draw conclusions about a *population parameter*

An experiment

Research question: How good are people at picking random numbers?

- People tend to be more predictable than random
- Are people predictable when they pick a random number?

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Let's perform a little experiment

- Imagine rolling a standard 6-sided die
- Enter the imagined outcome of your roll in the poll

Go to www.menti.com and use the code 1981 3822

Expectations

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- Each person chooses a number, say X
- If everyone follows the instructions, the probability P that X is anything other than a number between 1 and 6 is zero

$$P(X = n) = 0 \text{ for } n \neq 1, 2, \dots, 6$$

- If people are good at picking random numbers, the probability P that X is any one of the numbers 1 to 6 is equal

$$P(X = n) = P(X = k) \text{ for each } 1 \leq n, k \leq 6$$



Simulating the distribution



- To check whether people can pick random numbers well, we can start by exploring what truly random picks look like.
- We can use R to simulate picking random numbers

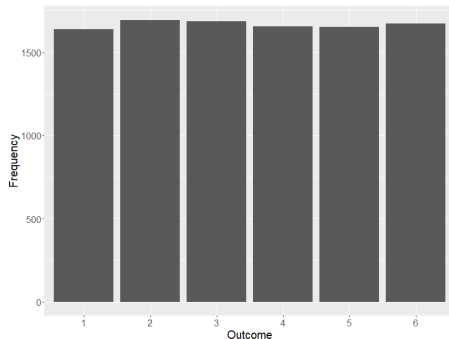
```
> ?sample  
> sample(1:6, 1)  
[1] 4
```



Simulating the distribution

Let's simulate the outcome of 10,000 random number picks

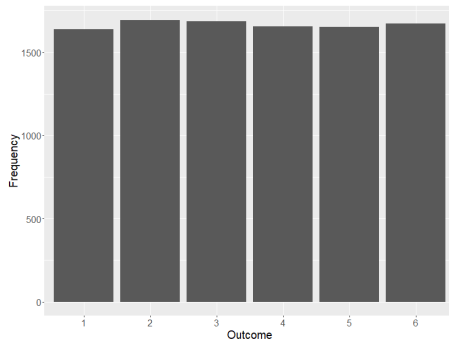
```
> randomPicks = replicate(10000, sample(1:6, 1))  
> ggplot(data.frame(randomPicks), aes(x=factor(randomPicks))) +  
  geom_bar() + xlab("Outcome") + ylab("Frequency")
```



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```



Of course, there aren't 10,000 respondents to our poll...

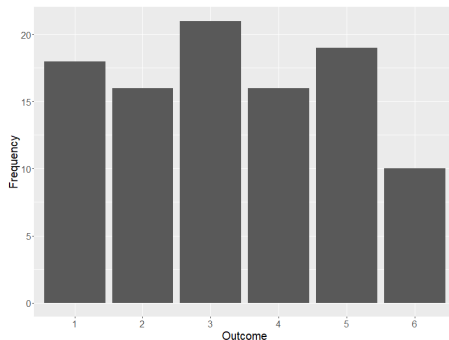


Simulating the distribution

Simulate the outcome of 100 random number picks



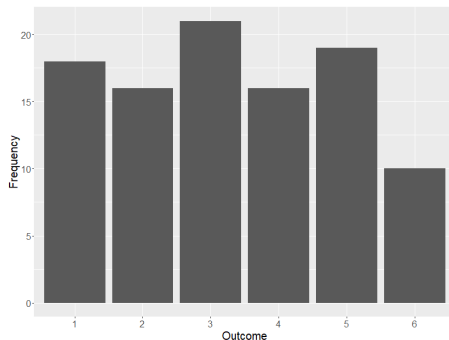
```
> randomPicks = replicate(100, sample(1:6, 1))  
> ggplot(data.frame(randomPicks), aes(x=factor(randomPicks))) +  
  geom_bar() + xlab("Outcome") + ylab("Frequency")
```



Simulating the distribution

Simulate the outcome of 100 random number picks

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> randomPicks = replicate(100, sample(1:6, 1))  
> ggplot(data.frame(randomPicks), aes(x=factor(randomPicks))) +  
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```



This doesn't really help to formulate expectations.



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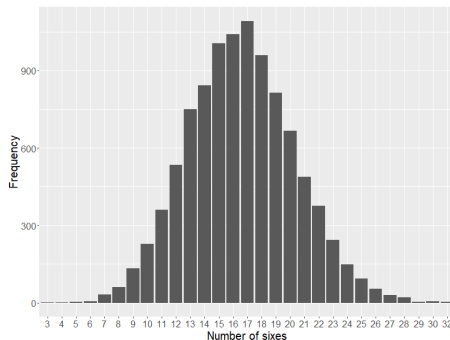
Maybe we should focus on one specific measurement.

- Which parameter though?

Simulating the number of sixes

Suppose we simulate our experiment and count the number of sixes

```
> sixCount = replicate(10000, sum(sample(1:6, n, r=T) == 6))  
> ggplot(data.frame(sixCount), aes(x=factor(sixCount))) +  
  geom_bar() + xlab("Number of sixes") + ylab("Frequency")
```

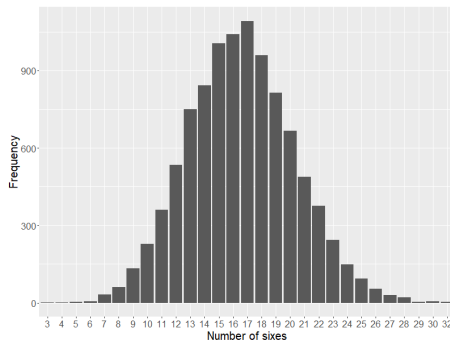


This could be helpful!

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This could be helpful! But 10,000 repetitions may not be enough...

Bernoulli random variable

Can we get the result of the simulation without actually doing the simulation?

Bernoulli random variable

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- A Bernoulli random variable X has 2 possible outcomes
 - A *success* $X = 1$ occurs with probability $P(X = 1) = p$
 - A *failure* $X = 0$ occurs with probability $P(X = 0) = 1 - p$
- “Choosing a six” can be modelled as a Bernoulli random variable with $p = 1/6$

Binomial distribution

A binomial distribution describes how likely it is to get k successes in n independent Bernoulli random variables, each with success probability p

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} = \frac{n!}{k!(n-k)!} p^k (1 - p)^{n-k}$$

Binomial distribution in R

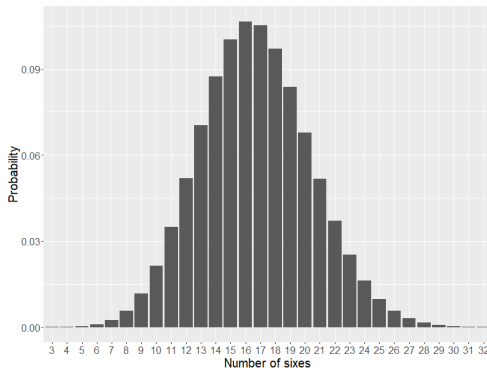
Rather than simulating, we can graph these probabilities directly



Binomial distribution in R

Rather than simulating, we can graph these probabilities directly

```
> binomData = data.frame(x=3:32, y=dbinom(3:32, 100, 1/6))  
> ggplot(binomData, aes(x=factor(x), y=y)) + geom_col() +  
  xlab("Number of sixes") + ylab("Probability")
```



Binomial distribution in R



Using the binomial distribution, we get data equivalent to a simulation of infinite size, without having to wait for that simulation to complete

```
> ?dbinom
> dbinom(16, 100, 1/6)      # P(X = 16 | n = 100, p = 1/6)
[1] 0.1065014
> pbinom(16, 100, 1/6)      # P(X <= 16 | n = 100, p = 1/6)
[1] 0.494159
> qbinom(0.25, 100, 1/6)    # Smallest x so P(X <= x | n = 100, p = 1/6) >= 0.25
[1] 14
> rbinom(10, 100, 1/6)      # Take 10 random outcomes
[1] 12 17 15 15 19 25 14 20 16 23
```



Pulling yourself up from your own bootstraps

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- What is the *actual* distribution of number of sixes in experiments like ours?

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 - I don't want to repeat the experiment 10,000 times...
 - But we do have our single sample...
 - Our sample is representative of the population...
 - Can we just take another sample from our sample?



Bootstrapping



A **bootstrap sample** samples from a dataset

- Sampling is done with replacement
- The resulting *replicate* has the same size as the original

Replicates can be used to describe the distribution of statistics

- These statistics are calculated for every replicate
- Bootstrapping typically involves at least 10,000 replicates



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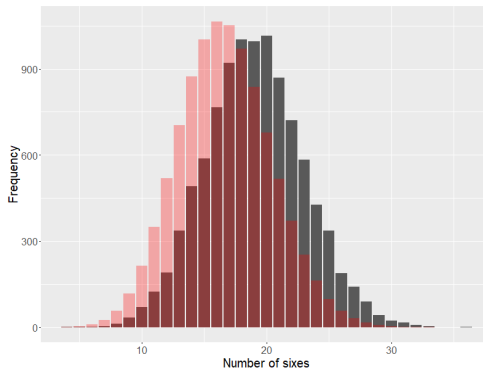
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Bootstrapping can be very useful

- Especially for complex statistics with unknown distributions
- Proportions and counts, however, have known distribution

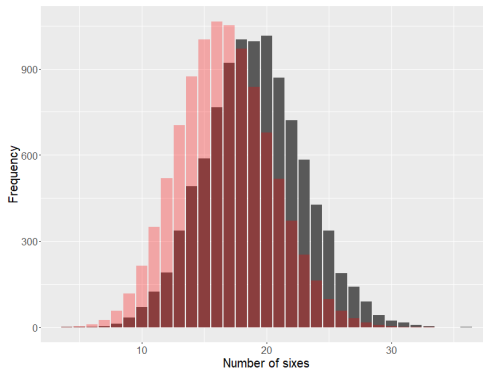
Example

```
> originalSample = sample(1:6, n, replace=TRUE)
> bootstrapSixCount = replicate(10000,
  sum(sample(originalSample, n, replace = TRUE) == 6))
```



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Bootstrapping results may not always be particularly accurate



Simulating the mean

Investigating the number of sixes can give some insight

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- It summarizes all data (unlike number of sixes)

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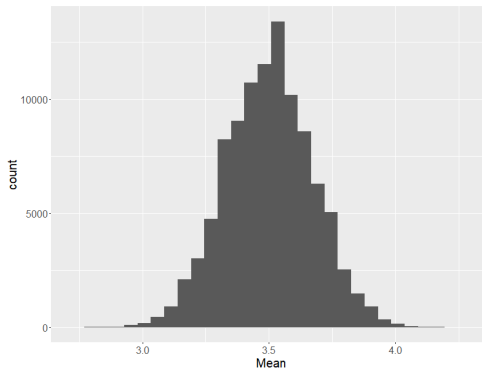
- It is a single number (unlike the histogram)
- It summarizes all data (unlike number of sixes)

Let's give the mean a try.

Simulating the mean

Let's simulate 100,000 experiments and calculate the means

```
> meanEst = replicate(100000, mean(sample(1:6, n, repl=TRUE)))  
> ggplot(data.frame(meanEst), aes(x=meanEst)) +  
  geom_histogram() + xlab("Mean")
```



Limitations

We run into a few limitations:

- The mean of 100 die throws is not quite categorical
 - There are likely over 100 unique means
- It's not quite continuous either
 - Some means are more common with combinations of the numbers 1-6 than others

Can we avoid these issues?

Central limit theorem

According to the **central limit theorem**, for a sufficiently large sample $X_1, X_2, \dots, X_n$ that are

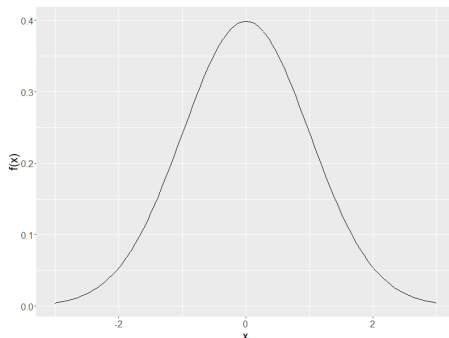
- *independent* and
- *identically distributed* with mean μ and standard deviation σ ,

the mean $\bar{X} = \frac{1}{n} \sum X_i$ is approximately normally distributed with mean μ and standard deviation $\sigma/\sqrt{n}$.

Normal distribution

The **normal distribution** is a continuous distribution described by

$$f(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

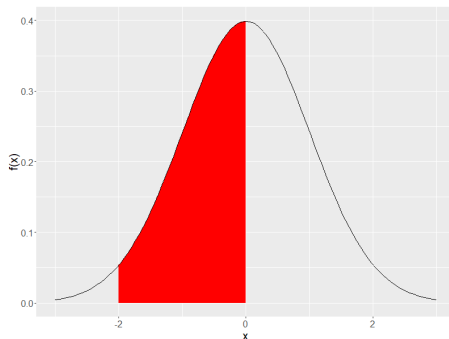


Normal distribution

$f(x)$ is density, **not** probability P

- For a continuous random variable X , $P(X = x) = 0$ for any x
- Probability is described by the *area under the graph*

$$P(X < x) = F(x|\mu, \sigma) = \int_{-\infty}^x f(x|\mu, \sigma) dx$$



Standard normal distribution

The **standard normal** distribution is a normal distribution with mean $\mu = 0$ and standard deviation $\sigma = 1$.

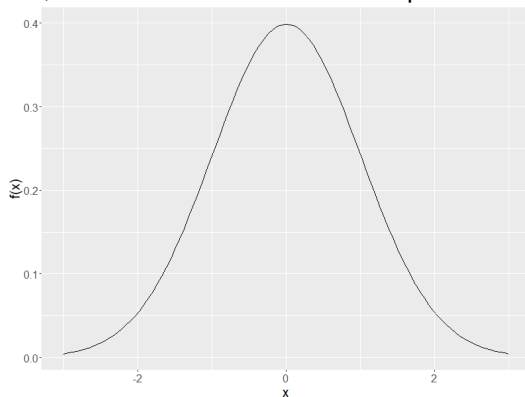
- When random variable X is normally distributed with mean μ and standard deviation σ , $Z = \frac{X - \mu}{\sigma}$ follows the standard normal distribution.
- In textbooks, Z is often used to refer to a random variable with a standard normal distribution

Normal distribution in R



Like the binomial distribution, the normal distribution is implemented in R

```
> dnorm(90, 100, 10)
[1] 0.02419707
> pnorm(90, 100, 10)
[1] 0.1586553
> qnorm(0.75)
[1] 0.6744898
> rnorm(1)
[1] -1.5780302
```



How much does size matter?

What is large enough?



- The central limit theorem holds for sufficiently large samples
- What is sufficiently large?

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- What is sufficiently large?
 - Let's find out



How much does size matter?

What is large enough?



- The central limit theorem holds for sufficiently large samples
- What is sufficiently large?
 - Let's find out
 - For normal-like distributions, a few is enough
 - For symmetric distributions, a few dozen maybe
 - For strongly skewed distributions, hundreds



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Applying CLT to our case

If people are truly choosing randomly, their mean choice should be approximately normally distributed with mean μ and standard deviation $\sigma/\sqrt{n}$

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How much does size matter?

Applying CLT to our case

If people are truly choosing randomly, their mean choice should be approximately normally distributed with mean μ and standard deviation $\sigma/\sqrt{n}$

- We know n , the number of responses
- What are μ and σ ?

How much does size matter?

The population mean

The population mean μ is the average value across the population

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 - E.g. one-sixth of the population will throw a six

How much does size matter?

The population mean

The population mean μ is the average value across the population

- Our population (all potential die throws) is infinite
- Still, we know the probabilities of all possible outcomes
 - E.g. one-sixth of the population will throw a six
- With this, we actually know enough to calculate the mean of an infinite population



How much does size matter?

The population mean

For a discrete random variable, the population mean

$$\mu = E(X) = \sum_k k \cdot P(X = k)$$

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So for our die throwing experiment, the population mean is

$$\begin{aligned}\mu &= E(X) = \sum_k k \cdot P(X = k) \\ &= \sum_k k \cdot \frac{1}{6} = \frac{1 + 2 + 3 + 4 + 5 + 6}{6}\end{aligned}$$

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$$\begin{aligned}\mu &= E(X) = \sum_k k \cdot P(X = k) \\ &= \sum_k k \cdot \frac{1}{6} = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5\end{aligned}$$

How much does size matter?

The population standard deviation

For a discrete random variable, the population variance

$$\sigma^2 = \text{var}(X) = E((X - \mu)^2) = \sum_k (k - \mu)^2 \cdot P(X = k)$$

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The population standard deviation

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$$\sigma^2 = \text{var}(X) = E((X - \mu)^2) = \sum_k (k - \mu)^2 \cdot P(X = k)$$

So for our die throwing experiment, the population variance is

$$\begin{aligned} \sigma^2 &= \text{var}(X) = E((X - \mu)^2) = \sum_k (k - \mu)^2 \cdot P(X = k) \\ &= \frac{(1 - 3.5)^2}{6} + \frac{(2 - 3.5)^2}{6} + \dots + \frac{(6 - 3.5)^2}{6} \end{aligned}$$

How much does size matter?

The population standard deviation

For a discrete random variable, the population variance

$$\sigma^2 = \text{var}(X) = E((X - \mu)^2) = \sum_k (k - \mu)^2 \cdot P(X = k)$$

So for our die throwing experiment, the population variance is

$$\begin{aligned} \sigma^2 &= \text{var}(X) = E((X - \mu)^2) = \sum_k (k - \mu)^2 \cdot P(X = k) \\ &= \frac{(1 - 3.5)^2}{6} + \frac{(2 - 3.5)^2}{6} + \dots + \frac{(6 - 3.5)^2}{6} = \frac{35}{12} \end{aligned}$$

Which makes the population standard deviation $\sigma = \sqrt{35/12}$.



How much does size matter?

Putting the CLT to the test

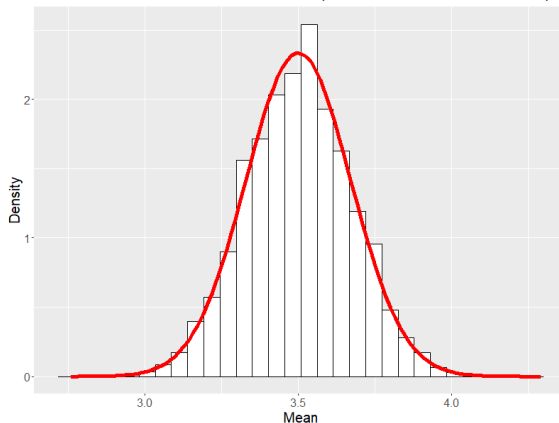
According to the central limit theorem, our simulated means should have a mean of $\mu = 3.5$ and a standard deviation of $\sigma/\sqrt{n} = \sqrt{35/12}/\sqrt{100} = \sqrt{35/1200}$.

How much does size matter?

Putting the CLT to the test

According to the central limit theorem, our simulated means should have a mean of $\mu = 3.5$ and a standard deviation of $\sigma/\sqrt{n} = \sqrt{35/12}/\sqrt{100} = \sqrt{35/1200}$.

```
> mean(meanEst)
[1] 3.500668
> sd(meanEst)
[1] 0.1704317
> sqrt(35/1200)
[1] 0.1707825
```



What about the standard deviation?

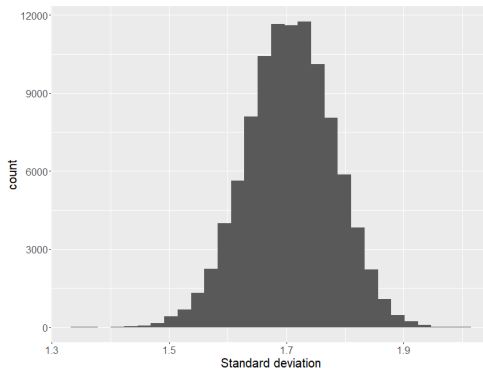
Rather than using the mean, we could use the standard deviation

$$s_x = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

Simulating the standard deviation

We simulate 100,000 experiments and calculate the standard deviations

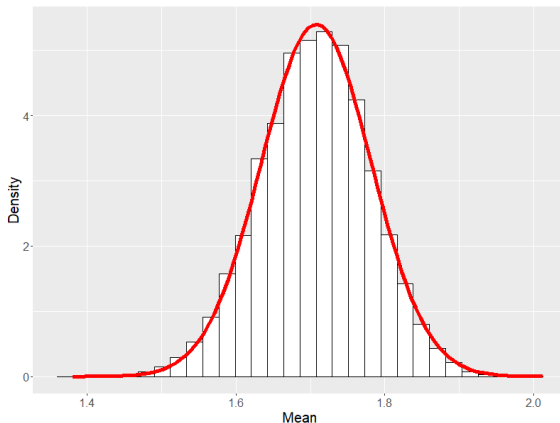
```
> sdEst = replicate(100000, sd(sample(1:6, n, repl=TRUE)))  
> ggplot(data.frame(sdEst), aes(x=sdEst)) +  
  geom_histogram() + xlab("Standard deviation")
```



Central limit theorem for standard deviation

According to the central limit theorem, our simulated standard deviations should have a mean of $\mu = \sqrt{35/12}$

```
> mean(sdEst)
[1] 1.705859
> sqrt(35/12)
[1] 1.707825
> sd(sdEst)
[1] 0.07388113
```



Summary

- With R, we can simulate experiment outcomes to get an impression of the distribution
 - Analyzing the sample distributions is more difficult than analyzing sample statistics
 - Bootstrapping can help uncover the distribution of sample statistics
- Bernoulli random variables have two possible outcomes
- Binomial distribution describes the probability of achieving k successful Bernoulli random variables out of n trials
- The mean of a sample is approximately normally distributed
 - Good approximation if distribution is symmetrical



Looking ahead

So far, we've looked at the relation between sample and population

- What conclusions about the population can we draw from a sample?
- How accurate are these conclusion?
- Are people good at picking random numbers?

Homework

Before the next class, you should

- read Chapters 6 and 7.1

Practical assignments

- a 3.14, 3.15, 3.16, 5.1, 5.6, 6.2, 6.12, 6.14, 7.1, 7.2
- b 6.15, 6.16, 6.17, 6.22, 6.25, 6.29, 6.30, 6.32

How random are dice?

The three-dimensional dynamics of the die throw

Kapitaniak, Strzalko, Grabski, and Kapitaniak; Chaos (22); 2012

“To summarize, the die throw is neither random nor chaotic.[...] We show that the probability, that the die lands on the face which is the lowest, is larger than on any other face, i.e., the dice are not fair by the dynamics.[...] If an experienced player can reproduce the initial conditions with small finite uncertainty, there is a good chance that the desired final state will be obtained.”

